

Lesson 5

Equipment: Raspberry Pi Python code editor, provided sheets, pens/pencils

Summary

This lesson will introduce pupils to Python's turtle module. The turtle module was chosen due to the similarity of its command to Scratch and pupils should be encouraged to think about similarities/differences between the two.

Lesson Aims

- I can identify variables in short programs including variables capturing user input
- I can identify where variables could be used in modelling real-life scenarios such as capturing different choices or characteristics
- I can use Python turtle to draw simple geometric shapes including a circle and a square/rectangle
- I can compare my experiences using Python turtle and Scratch

Curriculum Links

- TCH 2-13a: pupils should be able to structure a short program using Python turtle to draw a simple shape.
- TCH 2-14a: pupils should be able to use the following commands from Python's turtle library: `turtle.forward()`, `turtle.left()`, `turtle.right()` and `turtle.circle()`. Pupils are also encouraged to explore other commands related to colour and size of the pen.
- TCH 2-15a: pupils create their own shape programs using Python's turtle library.

Starter

- Prompt for pupils: compare the use of turtle in this lesson to what we saw in lesson 1 with Scratch pen. Which do you prefer and why?

Unplugged – Variables in various contexts (25 minutes)

- Pupils should complete, in order, the tasks detailed in the following sheets: TASK1 – PROGRAMS, TASK2 – GROUP DISCUSSION and TASK3 – SHAPES. For TASK3, pupils should be prompted to identify where variables could be used to model the given shapes. For example, factors such as shape, size and colour are acceptable responses here.
- The aim of this exercise is to make sure pupils are aware of different contexts where variables can be used. It should be emphasised that while pupils are encountering variables in different contexts, the underlying principle of variables as a being way of representing information that could have any of a set of values.

Taught Section (10 minutes)

- Pupils see examples of simple Python turtle usage including `.forward()`, `left()` and `right()`. Pupils should be shown how to use `turtle.circle()`. VPC should be used here as in lesson 1. One suggested method would be to initially show pupils some examples of using the forward and left/right commands, getting pupils to predict what these will do. Once pupils have some familiarity with these, a suggested activity would be to draw a square while asking pupils at each step what they think the next command should be, same as in lesson 1.

Coding Activity (30 minutes)

- Pupils are tasked to write code to draw a circle and another basic shape of their choice. Shapes involving only right angles are likely to be the easiest for pupils to work. For pupils looking for extra challenge, shapes such as triangles and hexagons could be a good suggestion.
- Pupils should first write their code on paper to follow the VPC process used in previous lessons.
- After pupils have written code on paper, they will transfer this to the Python editor.
- The suggested partition of time is 10 minutes writing code on paper and 20 minutes to write on the editor.

Closing (10 minutes)

- Pupils will see an example of variables being used in an online shopping order. This is similar to the activity in lesson 3 with the welcome messages, however will likely be more challenging. Answers to this activity could include: name, total, size, fit, order number, product, delivery date etc.
- Pupils should reflect on the differences between their experiences coding with Scratch compared to Python turtle. This can be held as a class discussion.

Spatial skills

- The abstraction of variables from the task sheets can be viewed as a spatial process. Encourage pupils to discuss these tasks and share their thoughts with other.
- Use VPC in taught section through providing examples and getting pupils to predict what will happen. This lesson in particular is a good opportunity to do this as turtle provides a clear visual reference for pupils to observe the execution of code.
- Ensure that pupils are following a VPC in writing their code. Code should first be written on paper before pupils move on to writing it on the Python editor.
- One method to ensure pupils have done this is to get all pupils to type in their code and run it simultaneously as a whole class. As this lesson uses turtle, pupils should be able to clearly see if what has been drawn is what they expected. Pupils will then be given time to make corrections to their code.
- As in previous lessons, ensure pupils are comparing their final code on their editor to their first draft on paper. If possible, allow time for pupils to work on this comparison towards the end of the lesson.

General teaching advice

- Pupils will be provided with turtle cheatsheets. Pupils should use these to help write their code (both on paper and only editor).